

When Safe Search Budgets Do Not Transfer Across Graph-ANNS Rebuilds: [Experiments & Analysis]

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Abstract

Rebuilding a graph index can change the search procedure on which an approximate nearest-neighbor search budget was calibrated. We study when a source-build budget retains its recall target on a rebuilt index, how this transfer should be audited, and what target-side recovery costs. Index-Conditioned Budget Audit (ICBA) organizes the analysis around build-conditioned risk, isolated selection and certification roles, and explicit search-cost accounting. In a SIFT-100K and Arxiv-Nomic-100K query-profile replay after 5% data refresh and rebuilding, direct reuse of a Tiered Conformal Pooling (TCP) history pool has empirical target failure rates of 6.69% and 7.24%. Target-selection recalibration reduces these rates to 1.79% and 2.04%, with mean distance-count reductions of 44.39% and 44.79% relative to the registered fixed endpoint. Cross-family and scale analyses extend the evidence beyond this recovery case. The central systems implication is to treat a rebuilt index as a new calibration environment: measure portability, separate observed recovery from a valid certificate, and account for the cost of obtaining target information.

Keywords

approximate nearest-neighbor search, graph indexes, index rebuilding, risk auditing, budget transfer

1 Introduction

Search quality depends on both the graph index and the budget used to explore it. The index determines which paths a query can traverse, while the budget controls how much of that search is explored. HNSW and Vamana make this tradeoff central to efficient approximate nearest-neighbor search [11, 13]. After rebuilding, a budget selected on the previous graph may execute on a different search structure. Preserving the parameter value does not establish that its recall behavior is preserved. This creates an operational question at the boundary between index maintenance and query serving: *when can an existing budget policy be reused, and when is target-side information worth acquiring?*

The question has two distinct settings. Rebuilding an unchanged vector collection can alter the graph through construction order or randomization. Rebuilding after a data refresh additionally changes the collection and potentially the nearest-neighbor ground truth. Prior work documents construction sensitivity and develops efficient update mechanisms [8, 12]. Adaptive search methods address another part of the problem: allocating work to individual queries using predicted recall or search cost [5, 10, 14]. These capabilities make budget reuse attractive, but its value still depends on the relationship between the source calibration environment and the target index.

Evaluating that relationship requires more than comparing average recall. A policy can attain a high mean while a subset of queries falls below the requested quality. We study the probability that a

query’s recall is below a specified threshold, separately from the expected fraction of missed neighbors. In the primary experiment, failure means $\text{Recall@10} < 0.95$: because recall is discrete, even one missing neighbor is a failure. A 5% risk target therefore asks that at least 95% of queries recover all ten neighbors. This event is stricter than mean recall of 0.95 and must remain explicit when interpreting external methods.

Our organizing insight is that budget portability belongs to a *source policy, target build, and query workload*. We call the resulting workflow *Index-Conditioned Budget Audit* (ICBA). It records what information determines each action, separates target selection from certification and evaluation, and distinguishes absolute target failure from additional failure relative to a reference. The theoretical analysis is organized around the observation assumptions under which source information can or cannot support a target decision. The empirical analysis asks how often reuse fails, which evidence supports recovery, and what that recovery costs. *Tiered Conformal Pooling* (TCP) provides the recovery case study: its history-pool replay reuses query-budget profiles and adjusts their decisions with target-selection information, followed by certification and fallback. The implemented first-passing pool and the assumptions needed for conformal coverage are distinguished in Section 4.2.

The primary replay covers SIFT-100K and Arxiv-Nomic-100K, with ten target builds per dataset after a 5% mixed delete/insert refresh. Direct TCP pool reuse yields empirical target failure rates of 6.69% and 7.24%. Target-selection recalibration reduces them to 1.79% and 2.04%, while reducing mean distance calculations by 44.39% and 44.79% relative to the registered fixed endpoint. These results concern queries with corresponding old-build profiles; they measure empirical recovery in that replay population. Search savings and a valid certificate for the entire candidate/fallback decision are separate outcomes.

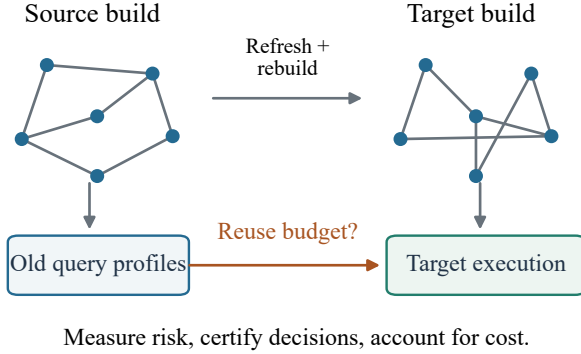
Figure 1 summarizes the operational question and the measured risk–cost tradeoff: direct reuse is cheap but exceeds the empirical risk target, whereas target recalibration retains a large part of the search savings.

The broader evidence is deliberately layered. External-method bridges examine query-level risk under their supported interfaces, while Vamana and Deep1M analyses examine different graph-family and scale conditions. These blocks retain their own references and failure definitions. Recovery is then evaluated against its acquisition cost: obtaining target labels and profiles can outweigh cheaper searches over a short serving horizon. Thus, an operator needs both a risk assessment and an amortization calculation before replacing direct reuse with recalibration.

We make four contributions:

- (1) A build-conditioned formulation that separates absolute risk, reference-relative transfer loss, and the information available to the budget policy.

(a) A budget policy is tied to an index version



(b) Observed risk–cost tradeoff

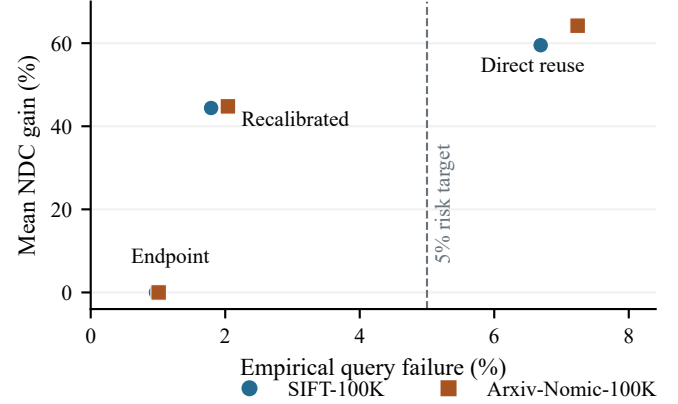


Figure 1: Budget portability and the primary replay. (a) Schematic, not a measured graph: after refresh and rebuilding, an old query-budget profile is applied to a changed index. (b) Frozen two-dataset outcomes, averaged over ten target builds per dataset. Gain is relative to each dataset’s fixed endpoint; recalibration includes fallback. The dashed line is the 5% empirical risk target, not a confidence bound.

Terms at a glance. SIFT and Arxiv abbreviate SIFT-100K and Arxiv-Nomic-100K in tables and figures.

Term	Meaning in this paper
ICBA	Index-Conditioned Budget Audit: the measurement and decision workflow.
TCP	Tiered Conformal Pooling: the history-based policy family; the primary replay uses a nine-history maximum and target grid shift.
NDC	Number of distance calculations; search work, not elapsed time.
CP / UCB	Clopper–Pearson / upper confidence bound.
Transport risk	Target failure of a transferred source policy, $\mathcal{R}_G(\pi_s)$.
LOTO	Leave one target build out.
Endpoint	Largest registered action; its risk is measured, not assumed zero.

- (2) An ICBA audit workflow connecting query roles, discrete failure events, certification decisions, and search-cost accounting.
- (3) A layered empirical study of budget portability across the registered recovery, method-bridge, graph-family, and scale settings.
- (4) A TCP recovery case study that quantifies the search benefit of target information and its workload-dependent acquisition cost.

Together, these contributions support treating the search policy as part of the index’s versioned operational state, with an explicit decision about reuse after rebuilding.

2 Related Work and Positioning

Graph construction, evaluation, and maintenance. HNSW combines hierarchical navigation with graph search, while DiskANN

introduces Vamana as part of a system for large-scale search [11, 13]. Azizi et al. [4] compare graph-search methods and construction paradigms across datasets and scales. Elliott and Clark [8] show that insertion order and intrinsic dimensionality affect HNSW recall. These studies establish that the constructed graph matters to search quality. We examine a different experimental unit: a budget policy transferred from a source build to a target build, with a matched reference used to separate target difficulty from transfer loss. FreshDiskANN [12] supports insertions and deletions without repeatedly reconstructing the entire index. Such maintenance mechanisms motivate distinguishing graph-only rebuilding from refresh-plus-rebuild; the latter is the setting of our primary TCP replay.

Adaptive work allocation and declarative recall. LAET predicts query-specific termination conditions using both static and intermediate-search features [10]. DARTH instead predicts achieved recall during execution and adapts prediction intervals to terminate HNSW or IVF searches at a declarative target [5]. ANNiE estimates the work required to reach a recall target from graph-derived features; its ANNiE-S search uses those estimates directly [14]. Ada-ef models query-to-dataset similarity distributions to choose an HNSW exploration parameter [16]. It also studies incremental updates to its statistics and estimation table, including comparisons with stale and fully recomputed variants after insertions or deletions. Our question concerns the validity and cost of carrying a decision between builds. It applies to adaptive policies as well as fixed budgets, but comparisons must retain each policy’s permitted signals and update procedure. The frozen DARTH and Ada-ef bridges are evidence about those integrations, rather than a common ranking of all published methods.

Quality guarantees and their target quantities. ConANN applies conformal risk control to IVF search, controlling expected false-negative rate while adapting the number of probed clusters [9]. Its

objective is the expected fraction of missed neighbors, whereas our principal event is whether a query falls below a recall threshold. ANNIe’s optional recall-target guarantee addresses the fraction of queries reaching the requested recall using quantile-loss minimization [14]. The DARTH+ preprint also introduces optional probabilistic quality guarantees for HNSW and IVF [6]. We study the build and policy covered by such an assessment, the information reused or reacquired, and the resulting behavior after rebuilding. The risk quantities remain explicit in comparisons.

Calibration, selection, and the role of ICBA. Conformal risk control bounds an expected monotone loss under its calibration assumptions [3]. Learn then Test formulates risk-controlled parameter selection as multiple hypothesis testing [2]. These are statistical foundations for auditing a search decision, rather than guarantees that automatically survive a changed execution environment. ConANN explicitly aligns calibration with index construction and assumes exchangeable query data [9]. In our setting, rebuilding can change the loss induced by the search even when the query population is retained. ICBA makes the target build, failure event, candidate selection, and fallback rule explicit. TCP supplies a concrete query-profile reuse and recovery case within that audit. This separation lets us ask whether target information restores useful search efficiency without conflating empirical recovery, a pipeline-level certificate, and net lifecycle savings.

3 Problem Formulation and Theoretical Lens

Budget transfer has three different statistical objects: a decision made from available observations, query risk on a fixed target build, and coverage over a distribution of builds. We state their guarantees separately. The testing, confidence-bound, and rank arguments below are classical tools applied to the registered search setting, rather than new general statistical inequalities.

3.1 Build-Conditioned Risk and Budget Labels

Let G include the vector snapshot, graph, and fixed search implementation, $q \sim \mathcal{Q}$ a query, and $\mathcal{A} = (a_1 < \dots < a_J)$ a finite grid of implementation-native actions. A policy π maps permitted observations to an action. For recall target τ , define

$$\begin{aligned} Z_G(q, a) &= \mathbf{1}\{\text{Recall}@k(G, q, a) < \tau\}, \\ \mathcal{R}_G(\pi) &= \mathbb{E}_{q \sim \mathcal{Q}} Z_G(q, \pi(q)). \end{aligned} \quad (1)$$

The arguments describing observations are suppressed in $\pi(q)$, but remain part of the policy definition. Ground truth is computed against the corresponding vector snapshot. In the primary replay, $k = 10$ and $\tau = 0.95$ require all ten neighbors; recovering nine is a failure.

For a source policy π_s , *transport risk* is its absolute target failure probability $\mathcal{R}_{G_t}(\pi_s)$ under the block’s declared failure event. Its empirical estimate is the fraction of target evaluation queries that fail when the source policy’s actions are executed. This is distinct from both source-build risk and the reference-relative increment. For a specified target reference π_t^{ref} , the incremental risk is

$$\Delta_{s \rightarrow t} = \mathcal{R}_{G_t}(\pi_s) - \mathcal{R}_{G_t}(\pi_t^{\text{ref}}). \quad (2)$$

Both absolute risks accompany this difference. A target reference can itself fail, and a refresh-plus-rebuild comparison changes

more than the graph alone. Requested action, distance evaluations (NDC), and elapsed time are recorded separately: ordering a_j does not order either realized search work or raw recall.

This distinction also affects budget labels. Define the first passing action $B_G^{\text{first}}(q) = \min\{a_j : Z_G(q, a_j) = 0\}$ and the stable-tail action

$$B_G^{\text{tail}}(q) = \min\{a_j : Z_G(q, a_\ell) = 0 \ \forall \ell \geq j\}. \quad (3)$$

An empty set has value $+\infty$, denoting an unresolved label, not an executable action. A finite stable-tail label establishes success of every grid action above it *on that recorded response*; a first passing label has no such implication when recall is non-monotone. Both are retrospective labels requiring truth. An under-budget event or a censored label is therefore kept distinct from the execution failure in Equation (1).

3.2 What Limited Observations Cannot Resolve

Fix a query and two possible build environments E_0, E_1 . Let O contain everything a decision rule can observe: source profiles, permitted query features, and any target probe transcript. Its laws are P_0^O, P_1^O . An action may encode execution, fallback, or abstention. Let D_i be the acceptable decisions in E_i under a declared risk–cost requirement, and let L_i be a nonnegative loss. Including cost matters: a common expensive endpoint may satisfy safety in both environments while failing the efficiency requirement.

PROPOSITION 3.1 (OBSERVATION-LIMITED DECISION LOSS). *If $D_0 \cap D_1 = \emptyset$ and $L_i(a) \geq \lambda \mathbf{1}\{a \notin D_i\}$ for $\lambda > 0$, every common, possibly randomized, decision rule based on O satisfies*

$$\max_i \mathbb{E}_i L_i(A) \geq \frac{\lambda}{2} (1 - \text{TV}(P_0^O, P_1^O)). \quad (4)$$

PROOF. Use the decision to test the environment: return 0 if $A \in D_0$, and 1 otherwise. Under E_0 a testing error implies $A \notin D_0$; under E_1 it implies $A \in D_0 \subseteq D_1^c$. The sum of expected losses is at least λ times the sum of testing errors, which is at least $1 - \text{TV}(P_0^O, P_1^O)$ by the two-point testing bound [15]. Taking half the sum proves the claim. $\square$

The result identifies an information requirement for avoiding conflicting decisions. It gives no positive lower bound if a common acceptable decision exists or the full observations distinguish the environments. For random queries, the relevant observable includes q jointly with the transcript; a bound on transcript marginals alone is insufficient. Our empirical transfer rates are measured outcomes, not numerical instantiations of Equation (4). In particular, classifier accuracy supplies a lower bound on distinguishability, whereas this loss bound needs an upper bound.

3.3 Fixed-Target Selection and Certification

Fix a target G and a family of M candidate policies. Write $r_j = \mathcal{R}_G(\pi_j)$ and $c_j = \mathbb{E} \text{NDC}_G(q, \pi_j(q))$. Selection estimates $\hat{r}_j, \hat{c}_j$ on its own query role. A useful action must be both distinguishable from unsafe candidates and separated from the risk boundary.

PROPOSITION 3.2 (MARGIN-BASED SELECTION). *On the event $\max_j |\hat{r}_j - r_j| \leq \varepsilon_R$ and $\max_j |\hat{c}_j - c_j| \leq \varepsilon_C$, retain $S = \{j : \hat{r}_j + \varepsilon_R \leq \delta\}$*

and choose $\hat{j} \in \arg \min_{j \in S} \hat{c}_j$. Every retained policy is safe. If j_* minimizes cost among policies with $r_j \leq \delta - 2\epsilon_R$ and this set is nonempty, then

$$r_{\hat{j}} \leq \delta, \quad c_{\hat{j}} \leq c_{j_*} + 2\epsilon_C. \quad (5)$$

PROOF. For retained j , $r_j \leq \hat{r}_j + \epsilon_R \leq \delta$. The margin gives $\hat{r}_{j_*} + \epsilon_R \leq r_{j_*} + 2\epsilon_R \leq \delta$, so $j_* \in S$. Thus $c_{\hat{j}} \leq \hat{c}_j + \epsilon_C \leq \hat{c}_{j_*} + \epsilon_C \leq c_{j_*} + 2\epsilon_C$. $\square$

This is a conditional selection statement relative to the margin-feasible comparator, not a bound against every safe policy. Its estimation event must be justified for the actual estimator. It also does not bound the cost of a fallback invoked after certification rejects the selected policy.

For a policy fixed independently of n i.i.d. target-certification queries, with x binary failures, a one-sided Clopper–Pearson bound [7] is

$$U(x, n; \alpha) = \begin{cases} \text{Beta}^{-1}(1 - \alpha; x + 1, n - x), & x < n, \\ 1, & x = n. \end{cases} \quad (6)$$

Acceptance requires $U \leq \delta$. Here α controls confidence error and δ bounds query failure. With zero failures, $U = 1 - \alpha^{1/n}$, so $\alpha = \delta = 0.05$ needs at least 59 certification queries. This minimum establishes possible acceptance, not high acceptance probability for a near-boundary policy.

PROPOSITION 3.3 (SELECTED-POLICY CERTIFICATE). *Freeze M policies independently of certification outcomes and allocate $\alpha_j > 0$ with $\sum_j \alpha_j \leq \alpha$. Compute $U_j = U(x_j, n; \alpha_j)$ on a common i.i.d. certification sample. Select any policy with $U_j \leq \delta$, or abstain if none qualifies. Then*

$$\Pr\{\text{a policy is selected and } r_{\text{selected}} > \delta\} \leq \alpha. \quad (7)$$

PROOF. The probability that any $r_j > U_j$ is at most $\sum_j \alpha_j \leq \alpha$. On the complementary event, every accepted policy is safe, regardless of which is selected. Independence among the policy bounds is unnecessary. $\square$

This is the standard simultaneous-testing construction [2]. Target selection may define the candidate family before certification; the statement then applies conditionally on that selection. A fallback belongs in the family or needs a separately justified certificate. The bound controls the probability of an unsafe selection, not the error rate conditional on the procedure selecting something, and it concerns one fixed target rather than an entire build campaign.

3.4 Pooling Across Exchangeable Builds

Historical query profiles provide a different route: pool budget labels across builds rather than certify a policy over target queries. Let m source builds and one target have exchangeable labels $B_1(q), \dots, B_m(q), B_t(q)$ for a fixed query q , with $+\infty$ retained in the order. For a build-marginal error level η , set $r = \lceil (1 - \eta)(m + 1) \rceil$. If $r \leq m$, let $a(q)$ be the r th source order statistic; otherwise abstain. An infinite order statistic also causes abstention.

PROPOSITION 3.4 (BUILD-MARGINAL LABEL COVERAGE). *For $r \leq m$, the preceding rule satisfies*

$$\Pr\{a(q) < \infty, B_t(q) > a(q)\} \leq \frac{m + 1 - r}{m + 1} \leq \eta. \quad (8)$$

If these are stable-tail labels, the same bound holds for finite-action execution failure.

PROOF. Break label ties using independent continuous auxiliary variables. Exchangeability makes the target’s augmented rank uniform among $m + 1$ positions. A label strictly larger than the r th source label has augmented rank greater than r , giving the inequality. For stable-tail labels, a finite action at least $B_t(q)$ succeeds by Equation (3). $\square$

The argument is conformal rank coverage with builds as the sampling unit [1]. It holds separately for each fixed query, marginally over the source and target build draw, not simultaneously for all queries or conditionally on a particular target. Nine sources have resolution 1/10; a non-vacuous 5% bound requires at least 19. Old-snapshot and refreshed-snapshot labels need not be exchangeable. First-passing labels additionally require a deployment-specific monotonicity bridge; averaging non-monotonicity over all grid actions does not supply that bridge for a selected action. These distinctions determine which parts of the theory can be used to interpret the TCP replay.

4 ICBA Workflow and the TCP Recovery Case

ICBA connects a portability measurement to an operational decision. It leaves graph construction and native traversal unchanged. Its inputs are source and target build identities, a query workload, the permitted policy observations, an action grid, a failure event, and a cost model. Its outputs separate direct-reuse behavior, the selected policy, certification status, fallback behavior, and lifecycle cost. TCP is the history-based budget policy used to study recovery within this workflow.

4.1 From a Transfer Measurement to a Decision

An audit first fixes the execution contract: vector snapshot, ground truth, implementation, grid, recall target, and endpoint treatment. The same parameter name across implementations is not a shared unit of work. We therefore retain native actions alongside NDC, recall, and measured timing, and identify unresolved labels before summarizing minimum budgets.

Target information then follows three paths. Selection chooses a candidate using its assigned queries. Certification checks the frozen candidate; it cannot retrain or reorder that candidate using evaluation outcomes. Evaluation records the completed decision’s risk and cost. The primary replay has disjoint roles of 500 selection, 500 certification, and 1,000 evaluation queries. Role separation concerns target responses: each query can still require a corresponding source-history profile, whose acquisition must be reported.

The audit preserves a direct-reuse row even when it would fail certification. Otherwise fallback could conceal the transfer failure being measured. For recovery rows, it reports the actual chosen policy and includes all fallback queries in risk and cost denominators. A certificate rejection is a statement about available evidence, not proof that the policy’s true risk exceeds δ .

4.2 The Frozen TCP Replay Policy

The primary refresh experiment uses the action grid

$$\mathcal{A} = (10, 20, 40, 80, 120, 160, 200) \quad (9)$$

for native HNSW search, with $\tau = 0.95$. For each query and old build, the implementation finds the first grid index meeting the recall target. It stores index J if no action passes. Write this encoded index as $\tilde{j}_g(q)$; its unresolved status is a separate property of the underlying responses. For target seed t , the history-max rule is

$$j_0^{(t)}(q) = \max_{g \in \mathcal{H}_t} \tilde{j}_g(q), \quad \pi_0^{(t)}(q) = a_{j_0^{(t)}(q)}, \quad (10)$$

where $\mathcal{H}_t$ contains the nine old builds other than the target-seed build. Executing the endpoint for an unresolved source query does not make its source label finite-safe; target outcomes are still measured normally.

This rule uses historical responses for the *same query*, including queries assigned to target evaluation. The experiment thus evaluates query-profile reuse after refresh, not prediction of a budget for an unprofiled query. No learned query-to-budget predictor is used in this lane. A fresh workload requires obtaining those profiles or declaring another action for a cache miss.

The historical TCP-HM9-TC variant instead uses stable-tail labels at $\tau = 0.90$, retaining $+\infty$ before routing unresolved queries to the endpoint. We keep that variant distinct from the first-passing, $\tau = 0.95$ refresh replay. Neither a shared name nor nine-source maximum pooling supplies the 5% build-marginal guarantee of Proposition 3.4; the label bridge, pool size, and exchangeability assumptions must each hold.

4.3 Target Recalibration and Fallback

Recalibration changes one global grid shift, not a per-query target label lookup. For $s \in \{0, \dots, J-1\}$, define

$$\pi_s(q) = a_{\min\{j_0^{(s)}(q)+s, J\}}. \quad (11)$$

The selection routine scans shifts in ascending order and chooses the first whose selection-role CP upper bound is at most 0.05. If none passes, it returns shift $J-1$, the endpoint policy. This use of the bound is a selection heuristic, not a simultaneous certificate over scanned shifts. It chooses the first qualifying shift, not the minimum measured-NDC policy of Proposition 3.2; larger shifts need not have monotonically increasing NDC.

The chosen policy π_s is frozen before certification. On the separate certification role, the code computes its individual 95% CP bound and the endpoint's individual 95% bound. It executes π_s if the former passes; otherwise it executes the endpoint and sets a deployment flag according to the latter. If both fail, the replay still records endpoint outcomes with a failed deployment flag. Such a record is evaluation evidence, not an authorized deployment. The source-audited comparator follows the same decision path with $s = 0$; raw source reuse bypasses fallback for diagnostic purposes.

These two individual bounds do not instantiate Proposition 3.3 at $\alpha = 0.05$: their error allocations sum to 0.10. The primary tables therefore retain empirical risk and fallback behavior without a pipeline-wide 95% claim. A separate historical three-policy implementation allocates 0.05/3 to TCP, a source-selected fixed action,

and the endpoint. That composition matches the simultaneous construction, but uses a different policy/event configuration and does not retroactively certify the refresh results.

Figure 2 makes the primary replay's information flow explicit. Selection determines the global shift; certification determines candidate acceptance and the fallback flag. Evaluation measures the completed decision, with no return path to selection.

4.4 What the Operator Pays For

Given a stored pooled index, applying Equation (11) costs a lookup, integer addition, and grid clipping per query before native search. Pooling m stored labels costs $O(m)$; retaining all labels for Q profiled queries costs $O(mQ)$ storage, or $O(Q)$ for an already pooled policy. These operation counts exclude acquiring the labels. Source profiling requires native searches and exact-neighbor truth, while target selection and certification require their own labeled responses. A first-passing profile can stop at a success; a stable-tail profile needs the remaining registered responses too.

The serving gain is the reduction in total NDC relative to the fixed endpoint over the same queries, not the mean of per-query ratios. The audit records mean, query-pooled p95/p99, and fallback-build counts separately, since a build-level endpoint fallback can change the entire cost distribution. Section 8 accounts for the horizon needed to repay information acquisition. An economical policy must repay those costs under its actual query-reuse rate; a cheap lookup alone establishes no lifecycle advantage.

5 Experimental Protocol

Primary refresh experiment. The primary workload has 100,000 base vectors per dataset: 128-dimensional SIFT-100K with Euclidean distance and 768-dimensional Arxiv-Nomic-100K with normalized inner product. A seed-991 membership change replaces 5% of the original collection with previously excluded vectors, preserving cardinality. Exact neighbors are computed against the refreshed snapshot. We reuse ten registered old-snapshot and ten refreshed-snapshot indexes per dataset. The native replay runs the official DARTH Faiss-HNSW fork, with $M = 16$, construction $ef = 100$, one search thread, $k = 10$, and the seven-action grid in Equation (9). Thus the primary recovery experiment is a Faiss-HNSW replay; the hnsplib scale evidence is a separate block.

For each target, 500 queries select a policy, a disjoint 500 certify it, and a disjoint 1,000 evaluate it. The same role IDs are reused across builds. Content-hash checks separate these roles from base vectors, insertion reserves, and historical query roles. The source-history profiles in Section 4 exist for these same queries on the old builds. This is a recurring-profile workload after refresh plus rebuilding; it isolates neither a graph-only causal effect nor generalization to unprofiled queries.

Metrics and decisions. We report empirical Recall @10 < 0.95 risk, mean distance evaluations, query-pooled p95 and p99 distance evaluations, candidate acceptance, and actual fallback. Relative search gain is

$$g(\pi) = 1 - \frac{\bar{c}_\pi}{\bar{c}_{\text{end}}}, \quad (12)$$

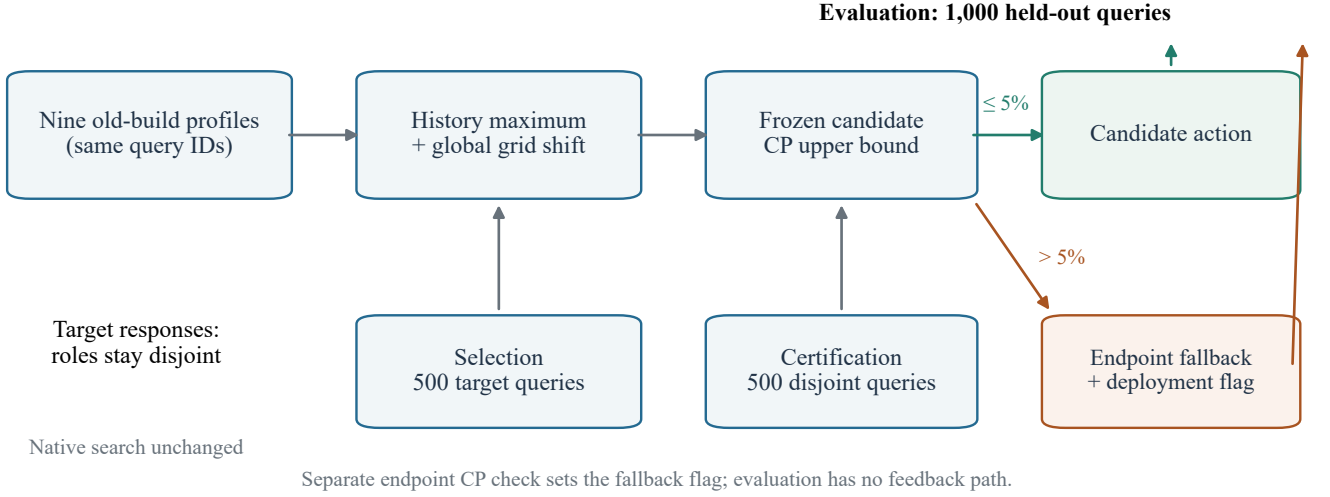


Figure 2: Executable ICBA/TCP replay workflow. Old-build profiles exist for the same query IDs in all three target roles. Only selection responses choose the shift. Independent certification checks the frozen candidate and the endpoint separately; if the candidate fails, the endpoint executes with its own deployment flag. A failed endpoint check remains an unqualified evaluation record. Individual CP checks here do not provide a joint 95% guarantee for the two-policy decision.

where $\bar{c}_{\text{end}}$ is the fixed- $ef = 200$ mean on the same evaluation queries and target builds. Fallback executions remain in all denominators. Candidate and endpoint checks use the individual one-sided 95% CP bounds specified in Section 4.3; the empirical decision and joint confidence accounting are reported separately.

Uncertainty and robustness. The registered nested bootstrap draws target builds with replacement, then draws paired policy/reference queries within each selected build (5,000 replicates, seed 991). Each replicate computes a ratio of pooled means. Intervals are percentile summaries of this procedure. Shared query IDs across builds introduce crossed dependence that independent within-build resampling does not fully preserve; the reported intervals retain that limitation. Leave-one-target-out (LOTO) and deletion of the largest relative-gain build provide direct sensitivity checks over the ten fixed targets. Observed p95/p99 reductions are descriptive comparisons, not hypothesis-test noninferiority certificates. NDC is the common search-work measure; none of these gains establishes an equal wall-clock speedup.

Evidence order. RQ1 and RQ4 establish direct-reuse behavior and its audit decision. RQ5–RQ8 examine recovery, tails, build sensitivity, and amortization. RQ2–RQ3 then test the scope of the finding through external-method, graph-family, and scale evidence. RQ9 closes the numerical and replay provenance. Each block keeps its native action, reference, and query-risk definition.

6 Portability and the Audit Decision

6.1 RQ1: Does the Old Budget Remain Safe?

Question. Does the old-build budget meet the 5% query-failure target after refresh and rebuilding? **Protocol.** We execute the nine-history TCP pool without target recalibration or fallback on each of the ten targets per dataset. **Observation.** Direct reuse saves 59.51%

Table 1: Primary 5%-refresh replay: empirical query risk and mean NDC gain relative to the fixed endpoint. “Audited reuse” includes endpoint fallback on every target.

Dataset	Policy	Risk (%)	Gain (%)
SIFT	Endpoint	0.97	0
	Direct reuse	6.69	59.51
	Audited reuse	0.97	0
	Target recalibration	1.79	44.39
Arxiv	Endpoint	1.01	0
	Direct reuse	7.24	64.21
	Audited reuse	1.01	0
	Target recalibration	2.04	44.79

of mean distance evaluations on SIFT-100K and 64.21% on Arxiv-Nomic-100K, but target failure is 6.69% and 7.24%, respectively. The fixed endpoint has risks 0.97% and 1.01%. **Conclusion.** Low serving cost does not make the reused policy safe under the registered target event. The available primary summary establishes failure of target qualification after the combined refresh/rebuild intervention; it does not quantify a source-to-target risk increase against a source-certified reference.

Failure concentration. With 1,000 evaluation queries per target, direct-reuse risk exceeds 5% on all ten builds of each dataset. Per-build risks span 5.70–7.40% on SIFT-100K and 6.40–8.20% on Arxiv-Nomic-100K. The largest-failure build contributes 11.06% and 11.33% of its dataset’s failures; deleting it leaves risks of 6.61% and 7.13%. Figure 3(b) shows that failure is distributed across targets, not concentrated in one outlier. This descriptive check addresses build concentration; identifying the causal graph-topology mechanism is outside this replay’s scope.

6.2 RQ4: What Decision Does ICBA Change?

Question. Does target auditing distinguish unsafe efficiency from a usable recovery action? **Protocol.** The audited-reuse comparator checks the unshifted policy; the recovery comparator first selects a global shift on selection queries. Both then use the separate certification role and registered fallback rule. **Observation.** Audited source reuse falls back on all ten targets of each dataset, retaining endpoint risk and zero serving savings. Recalibration accepts the candidate on ten SIFT-100K and nine Arxiv-Nomic-100K targets; the remaining target uses the endpoint. Figure 3(a) identifies each decision. The completed decisions have empirical risks 1.79% and 2.04%. **Conclusion.** The audit rules out direct reuse in these targets, while target selection recovers a lower-cost action in 19 of 20 cases. Candidate acceptance is an individual-CP outcome; its distinction from a joint 95% pipeline certificate follows Section 4.3.

7 Recovery Quality and Build Sensitivity

7.1 RQ5: Can Target Recalibration Recover Value?

Question. Can target-selection information recover search savings while meeting the empirical risk target? **Protocol.** The recovery lane selects one global upward shift over the frozen source pool on 500 queries, checks it on a separate 500, and evaluates the resulting candidate/fallback decision on 1,000. **Observation.** Relative to direct reuse, risk drops from 6.69% to 1.79% on SIFT-100K and from 7.24% to 2.04% on Arxiv-Nomic-100K. Mean NDC gains [95% interval], in percent, are 44.39 [43.88, 44.91] and 44.79 [34.72, 50.15], respectively. **Conclusion.** A target-selected global adjustment recovers empirical safety and substantial search savings on both recurring-profile workloads. It sacrifices some of direct reuse’s unsafe savings to reduce failures.

7.2 RQ6: Are Mean Gains Bought with a Worse Tail?

Question. Does the completed recovery decision increase high-percentile distance counts? **Protocol.** We compare pooled quantiles on identical evaluation queries, including the one endpoint-fallback target, against fixed-endpoint execution. **Observation.** Both p95 and p99 are lower on both datasets (Table 2). Fallback occurs on 0/10 SIFT-100K and 1/10 Arxiv-Nomic-100K targets. On the latter’s fallback target, the serving distribution equals that target’s endpoint distribution by construction. **Conclusion.** The observed pooled tails improve alongside the mean, without relying on widespread fallback. These pooled results support the registered descriptive tail criterion. They do not establish a separate nonfallback pooled-tail guarantee or a latency guarantee; per-build quantiles cannot be averaged to obtain either pooled quantile.

7.3 RQ7: Does One Build Explain the Gain?

Question. Does recovery depend on a single favorable target? **Protocol.** We retain the registered nested bootstrap and recompute pooled gain after each target is omitted and after the highest relative-gain target is removed. **Observation.** Minimum LOTO gains are 44.37% and 44.23%; delete-largest gains are 44.37% and

Table 2: Query-pooled distance-count tails, endpoint → completed target-recalibration decision. Quantiles use all ten targets, including fallback. Fractional counts arise from quantile interpolation.

Dataset	p95: endpoint → target	p99: endpoint → target
SIFT	2,615.00 → 2,349.05	2,855.00 → 2,774.00
Arxiv	3,252.05 → 2,974.10	3,504.01 → 3,398.00

Table 3: Mean NDC gain (%) and build-sensitivity checks for recovery. Intervals use the registered resampling scheme in Section 5.

Dataset	Gain [95% interval]	Min. LOTO	Delete largest
SIFT	44.39 [43.88, 44.91]	44.37	44.37
Arxiv	44.79 [34.72, 50.15]	44.23	44.23

44.23% (Table 3). Both full-sample gain intervals have positive lower endpoints. **Conclusion.** No single target explains the mean benefit within the ten registered builds. The wider Arxiv-Nomic-100K interval also makes the variability of its completed candidate/fallback decisions visible. Here, deleting the largest relative-gain target also gives the minimum LOTO gain, explaining the identical sensitivity columns.

8 RQ8: When Does Recovery Repay Its Cost?

Question. At what serving horizon does cheaper search repay the information used to obtain it? **Protocol.** The frozen ledger expresses cost in distance evaluations:

$$C_{\pi}(N) = C_{\text{acquire},\pi} + N\bar{c}_{\pi}, \quad (13)$$

$$N_{\text{BE}} = \frac{\Delta C_{\text{acquire}}}{\bar{c}_{\text{ref}} - \bar{c}_{\pi}}, \quad (14)$$

where $\Delta C_{\text{acquire}}$ is acquisition cost above the reference. There is no finite break-even if serving savings are nonpositive and this increment is positive. The recorded numerator includes old-build search profiles for the evaluation query set, all-grid target-selection searches, selected-candidate certification searches, and exact target truth for the 500 selection and 500 certification queries. Truth is modeled as a scan of 100,000 vectors per query. The graph rebuild is common to both choices. Control operations receive zero distance calls, which is not a zero-time assumption.

Observation. The dataset-level ratio of mean acquisition cost to mean serving savings gives 138,836 executions for SIFT-100K and 107,378 for Arxiv-Nomic-100K. All SIFT-100K targets have positive serving savings; one Arxiv-Nomic-100K target has zero savings and does not amortize. At $N = 10^6$, the modeled net reductions average 697,963,656.30 and 958,291,849.60 distance evaluations per target (Table 4).

Interpretation. The ledger does not separately charge old-profile truth, old-build profiling of selection/certification queries, or endpoint-certificate search. The reported break-even therefore applies when these omitted acquisitions are already available or

(a) Completed target decisions: 19 accepted candidates, 1 endpoint fallback

SIFT-100K	A	A	A	A	A	A	A	A	A	A
Arxiv-Nomic-100K	A	F	A	A	A	A	A	A	A	A
	1103	1229	1361	1499	1621	1747	1877	1999	2131	2267

A = accepted candidate F = fallback Columns: registered target-build seeds

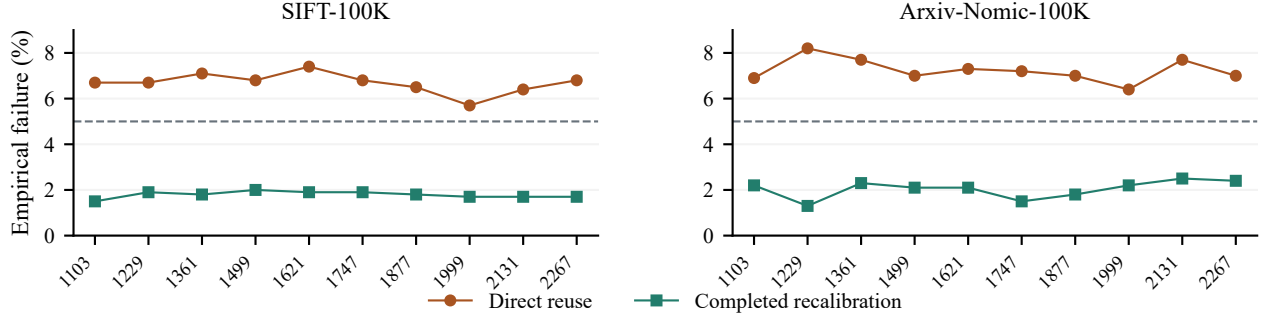
(b) Direct-reuse failures are spread across builds; recalibration reduces risk

Figure 3: Build-resolved audit outcomes from the frozen primary replay. (a) Each cell is one target: 19 candidates are accepted and Arxiv-Nomic-100K seed 1229 falls back. Its candidate CP upper bound is 5.056%, just above the fixed 5% threshold; the threshold is unchanged. All 20 completed decisions pass their applicable individual check. (b) Empirical evaluation risk on 1,000 queries per build. Direct reuse exceeds 5% on every target; completed recalibration remains below it. Points are descriptive rates, not confidence bounds; connecting lines indicate the fixed display order of seeds.

Table 4: Conditional lifecycle model. Break-even is a dataset-level ratio of means, not a guarantee for every build. The final column counts targets with no positive serving saving.

Dataset	Break-even	Net saving at 10^6	No saving
SIFT	138,836	697,963,656.30	0/10
Arxiv	107,378	958,291,849.60	1/10

cost-shared; otherwise it is optimistic. For an additional incremental cost $\Delta C \geq 0$, the corrected threshold is $N_{BE} + \Delta C / (\bar{c}_{end} - \bar{c}_\pi)$ whenever the denominator is positive. The actual query-reuse rate determines how often a new source profile must be acquired. Distance counts also exclude hardware-dependent overheads such as data movement, lookup time, and training or control time.

Conclusion. Recovery has positive long-horizon value under the reported acquisition model, while short-lived or nonamortizing targets favor the endpoint. Full deployment economics requires pricing the omitted information for the intended query workload; the serving gain alone is insufficient.

9 External Methods, Graph Family, and Scale

9.1 RQ2: Is Query-Risk Qualification Specific to TCP?

Question. Does the need to audit query-level risk extend to other adaptive search methods? **Protocol.** Frozen adapters run DARTH on both 100K datasets and Ada-ef on ten Arxiv-Nomic-100K targets, then evaluate the registered Recall@10 < 0.95 event. Their original observation and calibration interfaces are retained, so this is a bridge audit rather than an equal-information leaderboard. **Observation.** DARTH has empirical target risks 22.55% and 23.12%. Ada-ef’s target risk [95% interval], in percent, is 9.52 [9.44, 9.60] on Arxiv-Nomic-100K. Each exceeds the 5% query-failure threshold. **Conclusion.** Target qualification is consequential beyond TCP. These absolute risks alone do not identify a rebuild-induced increment: doing so requires matched source qualification and a target reference. Nor does this stricter event invalidate guarantees stated for mean recall or expected missed-neighbor fraction. The bridge outcome supports the audit’s use, not superiority over the published methods.

9.2 RQ3: Does Transfer Loss Extend across Scale and Family?

Question. Does the phenomenon appear beyond the primary 100K Faiss-HNSW recovery case? **Protocol.** The Deep1M block builds

Table 5: Protocol-separated extension evidence, reported as estimate [95% interval] when available. Risks are percentages; the Deep1M increment is in percentage points (pp). Intervals retain each block’s registered resampling unit and are not pooled across protocols.

Block	Observation and interpretation
DARTH	SIFT 22.55, Arxiv 23.12; absolute target qualification failure
Ada-ef	Arxiv 9.52 [9.44, 9.60]; absolute target risk
Deep1M	Additional 20.04 pp [17.29, 22.77] above reference 8.82
Vamana	SIFT 16.86 [15.88, 17.93]; Arxiv 11.37 [10.80, 11.99]; post-hoc realignment

eight hnsplib indexes, crossing four seeds with two insertion orders, and replays 56 directed source–target pairs on 1,000 queries. It compares source-derived actions with each target’s own first-passing action under the same discrete recall event. The separate Vamana realignment uses six source and six distinct target builds, 36 directions, and 750 queries; its native action is search-list size at beam width one, not HNSW *ef*. **Observation.** Deep1M transport risk is 28.86% against 8.82% target-reference risk. Its increment [95% interval], in percentage points, is 20.04 [17.29, 22.77]. The nonzero reference risk retains queries with unresolved labels within the grid. Vamana’s post-hoc unified analysis reports 16.86% and 11.37% transport risk (Table 5); its cost-tax intervals include zero. **Conclusion.** Deep1M supplies a matched additional-loss result at larger scale, rather than a safe-reference-to-unsafe-target transition. Vamana adds cross-family directional evidence from already observed outputs. Native action and censoring semantics remain block-specific, so neither extension is pooled into the primary recovery effect or a cross-engine cost ranking.

10 Reproducibility and Operational Implications

10.1 RQ9: Can the Numbers and Decisions Be Replayed?

Question. Can an evaluator trace the reported numbers to the inputs and decision rules? **Protocol.** The evidence package binds input hashes, role IDs, analysis scripts, per-build tables, and decision manifests to a frozen numerical ledger. Its checker selects unique CSV/JSON rows, verifies risk/gain aggregation, and regenerates the LaTeX value macros. The portable interface separates a lightweight smoke, table reconstruction, an external-asset pre-flight, and guarded full replay. **Observation.** The existing portability audit preserved hashes of all 80 pre-audit scientific files, passed a neutral-path smoke and 28 focused tests, and replayed the DARTH, Vamana, and Deep1M analyses from sealed inputs through explicit root arguments. The manuscript’s 57 frozen values and its additional tail/sensitivity macros are checked against committed summaries; the tables themselves are typeset from those macros. **Conclusion.** The package supports numerical traceability and tested replay interfaces. Recreating raw searches requires the separately provisioned vectors, truth, indexes, and native binaries identified by the checksum ledger. These reference-server checks are distinct from independent replication on a new installation.

Evidence and scope. The strongest recovery evidence is the two-dataset, ten-target recurring-profile experiment. The extension blocks answer narrower questions under their own reference policies. The theory separates observation-limited decisions, fixed-target query certificates, and build-marginal label coverage; the replay does not measure full-observation total variation or instantiate all conditions of the build-coverage result. These boundaries explain why empirical recovery, selected-policy confidence, and economic value are reported as separate outcomes.

Index-operation decision. An operator should version a budget policy together with its index snapshot, search implementation, query-profile population, and calibration record. A changed build triggers a concrete decision: audit reuse, acquire target evidence for recalibration, or retain the endpoint. Rejection can reflect inadequate certification power rather than actual unsafety. Conversely, an efficient policy that fails the registered event should not be hidden by reporting only its post-fallback average. The evidence favors selective recalibration when the profiled workload recurs long enough to repay acquisition, with a fresh cost and risk check for a changed workload.

Data and software availability. The reproducibility materials comprise protocol manifests, input checksum lists, analysis and table-generation code, compact result tables, and replay instructions. Large external vectors and pre-existing indexes are provisioned separately. The package distinguishes the lightweight numerical check from raw-query replay and identifies which inputs each requires.

11 Conclusion

Search-budget reuse across graph rebuilds is a measurable transfer decision. The frozen evidence shows target failures under direct source reuse and empirical recovery from target recalibration in the query-profile setting. ICBA connects those observations to explicit risk events, independent query roles, and lifecycle accounting. The resulting lesson is to evaluate portability and recovery together: retained search efficiency, statistical assurance, and the cost of acquiring target information describe different parts of the same operational decision.

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